Handling Exceptions

CS300: Programming II

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Tophat Q0

When is our first CS300 midterm exam?

→ Open Tophat to vote.

What is one study tip you have? Share with the person next to you!

Midterm #1

Next Wednesday at 5:45 pm -- rooms TBA!

Covers Modules 01-03.

Do I need to know all of the types of Exception? **No.**
Do I need to know how Exceptions work? **Yes.**

The Throwable family

```
Throwable
├── Error                       // the JVM is in trouble; not yours to fix
└── Exception
    ├── RuntimeException        // UNCHECKED
    │   ├── NullPointerException
    │   ├── ArrayIndexOutOfBoundsException
    │   └── IllegalArgumentException
    ├── IOException             // CHECKED
    ├── InterruptedException    // CHECKED
    └── ...                     // CHECKED
```

See the full list here. We'll work with them today!

Eclipse Exercise

Today is mostly a coding exercise, download our starter (and solution!) code from Canvas.

Implement the output from the "happy path".

Learning Objectives

1. Be **prepared** to take the midterm exam!
2. Be able to **write** a `try` / `catch` that recovers from bad input.
3. Be able to **trace** control flow through `try` / `catch` / `finally`.
4. Be able to **debug** an unreachable `catch` block.

Exception Handling

We can handle an exception with a `try` / `catch`.

Where we handle it? That's up to us!

→ Step through it in Java Visualizer.

Eclipse Exercise

Let's let `main` catch the exception.

Reminder: You can also throw an exception of your own, e.g. `throw new IllegalArgumentException("NO!")`

Uncaught Exceptions

```
try {
  int track = Integer.parseInt(input);
  return lengths[track];
} catch (NumberFormatException e) {
  return -1;
} finally {
  System.out.println("finally: always");
}
```

`catch` only catches exception of their corresponding type (or their children!)

Trace it in Java Visualizer

Catch blocks are tried top down

```
try {
  System.out.println(playlist[5]);
} catch (RuntimeException e) {
  System.out.println("generic");
} catch (ArrayIndexOutOfBoundsException e) {
  System.out.println("specific");     // will this ever print?
}
```

Eclipse refuses to compile this. Why?

Catch blocks are tried top down

```
try {
  System.out.println(playlist[5]);
} catch (ArrayIndexOutOfBoundsException e) {
  System.out.println("specific");     // narrow, so it gets the first look
} catch (RuntimeException e) {
  System.out.println("generic");      // the fallback
}
```

Catch narrow to wide `ArrayIndexOutOfBoundsException` is a type of `Runtime Exception`!

Run it in Java Visualizer

Or catch both at once

```
} catch (NullPointerException | ArithmeticException e) {
  System.out.println("bad data: " + e);
}
```

One block, two types, for when the response is identical. Note the single `|`.

Eclipse Exercise

Finish up our implementation.

Questions?